Radar Safety Issues – TinySAR

These calculations are based on the Australian Standard cited as the Radiation Protection Standard for Limiting Exposure to Radiofrequency Fields —100 kHz to 300 GHz (2021).

Given the scientific advances surrounding the effects of RF fields since 1998, ICNIRP revised its RF guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz) in March 2020 (ICNIRP 2020a). For effects below 100 kHz ICNIRP revised its guidelines for static (0 Hz) and low **frequency** (1 Hz to 100 kHz) fields in 2009 and 2010, respectively (ICNIRP 2009; 2010).
It is Australian Government Policy to implement international best practice and to adopt international standards where they exist and can be applied to the Australian regulatory environment. This Standard is based on the ICNIRP (2020) recommendations for RF fields (ICNIRP 2020a).

Mandatory limits on exposure to RF fields are based on substantiated health effects and are termed 'basic restrictions'. Protection against substantiated adverse health effects requires that these basic restrictions are not exceeded. Depending on frequency, the physical quantities used to specify the basic restrictions are **induced electric field (E_{ind})**, **specific energy absorption rate (SAR)**, **absorbed power density (S_{ab})**, **specific energy absorption (SA)** and **absorbed energy density (U_{ab})**.

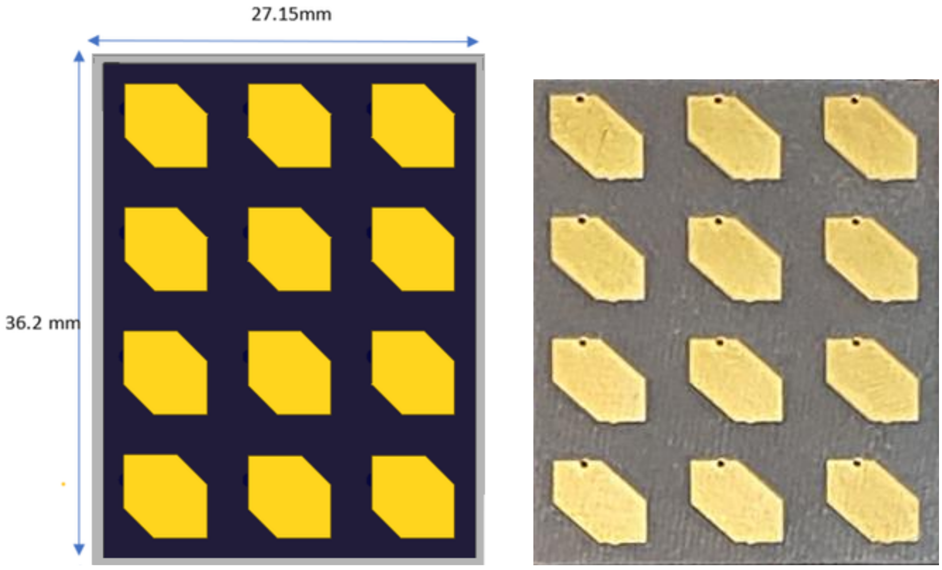
The maximum time averaged exposure limit that covers the 16.5 GHz band (6GHz to 300GHz) ranges from 100W/m^2 for occupational workers down to 20W/m^2 for the general public.

Table 1: Basic Restrictions for Incident Field Exposure for averaging intervals >= 6 minutes

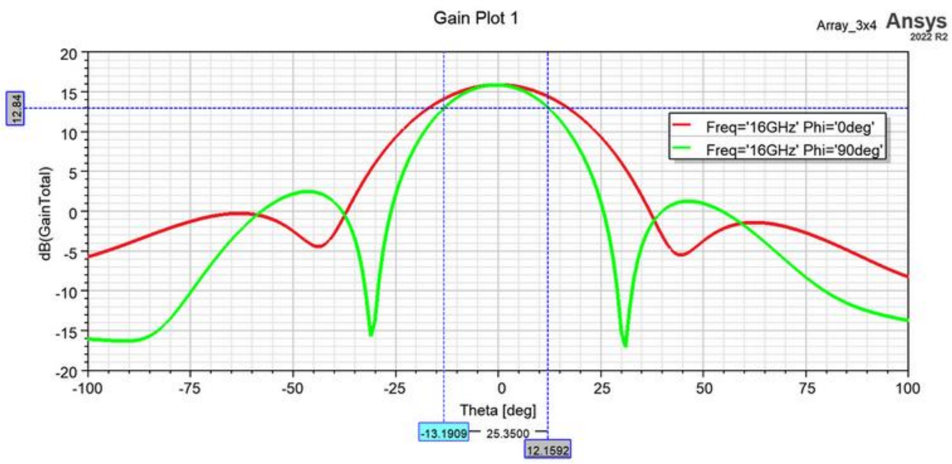
Exposure category	Frequency Range	Whole Body Average, SAR (W/kg)	Absorbed Power Density, S_{ab} (W/m^2)
Occupational	6GHz – 300GHz	0.4	100
General Public	6GHz – 300GHz	0.08	20

Single Sub-Array

The TinySAR radar being developed by Mission Systems for DAA and SAR operations from a drone can radiate a power of up to 0.5 W from the single sub-array shown in Figure 1.



(a)



(b)

Figure 1: Images (a) of the transceiver array and (b) the normalised azimuth and elevation beam patterns

The 3 dB beamwidth for the elevation (narrow) beam is approximately 25° and the azimuth (wide) beamwidth is 34°

The equivalent aperture d (m) can be calculated for a uniformly illuminated aperture using

$$d = \frac{58\lambda}{\theta_{3dB}} \tag{1}$$

where λ - wavelength (m)

θ_{3dB} – Beamwidth (deg)

This equates to 41.8 mm and 30.7 mm in the two cases. Note that this is slightly larger than the physical aperture, but the differences are insignificant.

As a rule of thumb, the transition between the near and far fields is defined at a range R (m) given by

$$R = \frac{0.5d^2}{\lambda} \tag{2}$$

Because the antenna has slightly different azimuth and elevation apertures, the near-far field transitions will also be different, 48.5 mm and 26 mm respectively. We will assume that the far field starts at (48.5+26)/2 = 37 mm for these calculations.

Case 1: A conservative estimate of the maximum power density W_m (W/m^2) in the near field (R < 37 mm) is given by

$$W_m = \frac{4P_T}{A} \tag{3}$$

where W_m – Maximum power density (W/m^2)
P_T – Net power delivered to the antenna (0.5 W)
A – Physical aperture area of the antenna (m^2)

For an area of 41.8 × 10^{-3} × 30.7 × 10^{-3} = 0.0013m^2, this equates to a maximum power density of 1559 W/m^2 if the object is within 37 mm of the antenna.

For the radar, the maximum power flux density equates to 1559 W/m^2. This is a factor of 78 times larger than the maximum continuous (> 6 minutes) average exposure limit for the general public. It is therefore advisable not to venture within the near field for extended periods. Shorter duration exposures are considered later in this report.

It is particularly important not to look into the radiating antenna at close range, even for short periods.

Case 2: Under normal conditions any exposure that occurs will be in the far field where the power density, W (W/m^2), is given by

$$W = \frac{P_T G}{4\pi r^2} \tag{4}$$

where G – Antenna gain with respect to Isotropic (ratio)
r – Distance from the antenna (m)

For a gain of 14 dB (25), and 0.5 W transmitter power, the power density equation reduces to

$$W = \frac{1}{r^2}$$

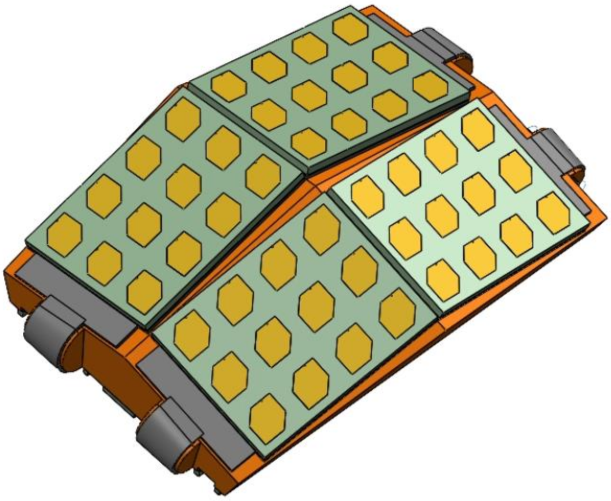
Table 2 shows the power density in the far field as a function of range

Table 2: Power Density in the Far Field for a Single Sub-Array Radiating 0.5 W

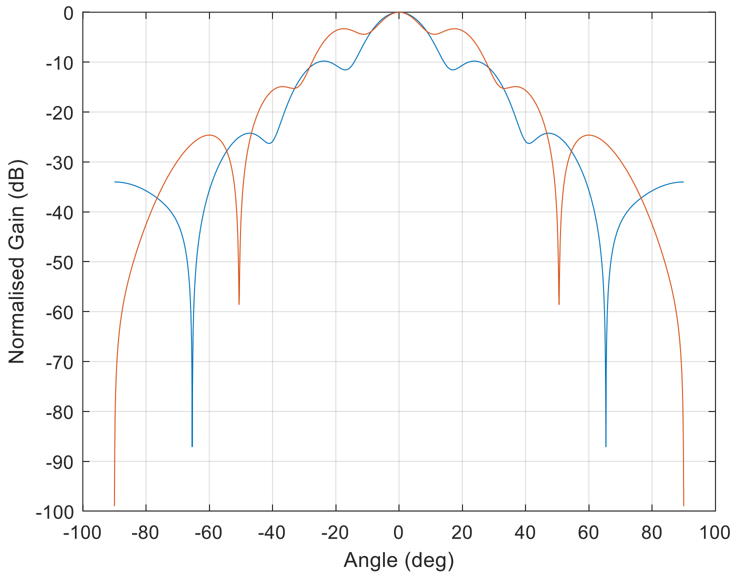
Range (m)	Power Flux Density ($W/m^{2)}$	Allowed Power Density (general public), S_{ab} (W/m^2)
0.01	100	20
0.22	20	20
2	0.25	20

For a single sub-array it is advisable to remain at least 220 mm from the radiating face of the antenna to comply with the safety requirements for the general public.

Cluster of 4 Sub-Arrays



(a)



(b)

Figure 2: Images (a) of the transceiver array cluster and (b) the normalised azimuth and elevation beam patterns

The 3 dB beamwidth for the elevation (narrow) beam is approximately 14.7° and the azimuth (wide) beamwidth is 15.5° with a directivity of 18.7 dB.

The equivalent aperture d (m) can be calculated for a uniformly illuminated aperture

$$d = \frac{58\lambda}{\theta_{3dB}} \tag{1}$$

where λ - wavelength (m)
θ_{3dB} – Beamwidth (deg)

This equates to 71 mm and 67 mm in the two cases.

As a rule of thumb, the transition between the near and far fields is defined at a range R (m) given by

$$R = \frac{0.5d^2}{\lambda} \tag{2}$$

Because the antenna has slightly different azimuth and elevation apertures, the near-far field transitions will also be different, 140 mm and 125 mm respectively. We will assume that the far field starts at (140+125)/2 = 133 mm for these calculations.

Case 1: A conservative estimate of the maximum power density W_m (W/m^2) in the near field (R < 133 mm) is given by

$$W_m = \frac{4P_T}{A} \tag{3}$$

where W_m – Maximum power density (W/m^2)
P_T – Net power delivered to the antenna (2 W)
A – Physical aperture area of the antenna (m^2)

For an area of 71 × 10^{-3} × 67 × 10^{-3} = 0.0048 m^2, this equates to a maximum power density of 1667 W/m^2 if the object is within 133 mm of the antenna.

For the radar, the maximum power flux density equates to 1667 W/m^2. This is a factor of 83 times larger than the maximum continuous (> 6 minutes) average exposure limit for the general public.

Case 2: Under normal conditions any exposure that occurs will be in the far field where the power density, W (W/m^2), is given by

$$W = \frac{P_T G}{4\pi r^2} \tag{4}$$

where G – Antenna gain with respect to Isotropic (ratio)
r – Distance from the antenna (m)

For a gain of 18.7 dB (74), and 2 W transmitter power, the power density reduces to

$$W = \frac{12}{r^2}$$

Table 3 shows the power density in the far field as a function of range.

Table 3: Power Density in the Far Field for a Cluster of Four Squinted Sub-Arrays Radiating 2 W

Range (m)	Power Flux Density (W/m^2)	Allowed Power Density (general public), S_{ab} (W/m^2)
0.5	48	20
0.77	20	20
5000	1	20

For a cluster of four sub-arrays each radiating 0.5 watt, it is advisable to remain at least 775 mm from the radiating face of the antenna to comply with the safety requirements for the general public.

In the case where the aperture is tapered by adjusting the gains of the individual elements, the antenna beamwidth will be increased, and the radiated power density, reduced. However, it is better to err on the side of caution, so we will stick to the longer range.

In the case where the sub-arrays are not squinted, the antenna gain and hence power density in the far field would be increased and the safety calculations would need to be reviewed.

Skin Depth

In addition, because of the high operational frequency, the skin effect limits the penetration of microwave radiation to the surface of the body.

$$\delta = \sqrt{\frac{2}{2\pi f \mu \sigma}} \text{ m}$$

Where δ - Skin depth (m)
f – Operational frequency (Hz)
μ - Permeability (H/m)
σ - Conductivity (mho/m)

For a non-magnetic medium the permeability is $4\pi x10^{-7}$, and the conductivity of human skin varies between 0.1 and 0.4 [3] which makes the skin depth something between 6 and 12 mm

Conclusions

The radar power density is significantly higher than the exposure thresholds for the general public in the near field for all antenna configurations. In the far field, the safe distance is 220 mm for a single sub-array radiating 0.5 W of power, and 775 mm for a cluster of four squinted sub-arrays radiating a total of 2 W.

Additionally, the penetration depth is small, so any minor effect will be constrained to the top 12 mm of the body.

That being said, it is still not as good idea to stare into any source of electromagnetic radiation, however low, for long periods of time.